

061006T6ICT

COMPUTER SCIENCE 6

IT/OS/ICT/CR/10/6

COMPUTER PROGRAMING

JAN/MARCH 2025



**KITALE NATIONAL POLYTECHNIC
(KNP INTERNAL EXAM)**

WRITTEN ASSESSMENT

Time: 2 Hours

INSTRUCTIONS TO CANDIDATES

1. This paper has TWO sections A and B. Attempt questions in each section as per instructions given in the section.
2. You are provided with a separate answer booklet.
3. Marks for each question are indicated in the brackets.
4. Do not write on the question paper

This paper consists of 4 printed pages

**Candidates should check the question paper to ascertain that all pages
are printed as indicated and that no questions are missing**

SECTION A: 40 MARKS

Answer ALL the questions in this section.

1. What is Programming? (2 Marks)
2. Compare and contrast C programming language to C++ programming. (6 Marks)
3. Mention FOUR examples of IDEs used in programming. (4 Marks)
4. Explain the difference between simple ***if*** and ***if else*** statements. (2 Marks)
5. Consider the following C++ code fragment:

```
#include<iostream>
using namespace std;
int x = 0, sum = 0;
while (x <= 5) {
    sum = sum + x;
    x++;
}
printf("The sum is: %d", sum);
```

Replace the while loop in the fragment above with a ***for loop*** that prints the same value of 'sum' variable. (4 Marks)

6. Performing program design and analysis is one of the critical stage in program development. Outline FOUR elements involved in program design and analysis. (4 Marks)
7. Outline THREE differences between *source code* and *machine code* as used in computer programming. (6 Marks)
8. A variable is essential in computer programming. State FIVE Rules when naming a variable. (5 Marks)
9. As an expert in the ICT field. State THREE software design tools. (3 Marks)
10. Identify FOUR advantages of using C to develop a computer program. (4 Marks)

SECTION B: 60 MARKS

Answer any THREE questions in this section.

11.

- a) IDE plays a crucial role in programming. Without the use of IDE most of the programmers will have difficulties in developing programs. Discuss any FIVE roles of IDE. (10 Marks)
- b) Peter a student at a technical institute was considering using a flow chart to design a program. Explain FIVE reasons why he could have considered this approach. (10 Marks)

12.

- a) Apart from enhancing readability of a program. Variable and function names in a program are essential for several reasons. Discuss FIVE reasons why a variable and function name are essential. (10 Marks)
- b) Discuss FIVE reasons why a beginner in programming is encouraged to learn C programming as their first language. (10 Marks)

13.

- a) Joseph a student at tusome Pamoja College didn't see the reason why a maintenance schedule should exist once the program has been developed. Explain FIVE reasons why you may differ with his reasoning. (10 Marks)
- b) Understanding the phases involved in developing a program is key in computer programming. Discuss FIVE phases involved in computer programming. (10 Marks)

14.

- a) Peter while developing a program in class, included comments. Discuss FIVE reasons for including his comments. (10 Marks)
- b) Mary a student at Igonje Technical training institute wants to understand the different escape sequences in C programming. Discuss FIVE escape sequences you will take her through. (10 Marks)